Depth Map Scene Classification: Are Depth-Pretrained Features Better Than Generic Self-Supervised Features?

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Abstract—Scene classification from depth maps is an important task for indoor robotics and privacy-preserving vision systems. A natural intuition is that backbones pre-trained on large-scale depth data—such as Depth Anything V2—would produce superior features for depth-based tasks compared to generic self-supervised models like DINOv2. In this paper, we challenge this assumption through controlled ablation experiments on the NYU Depth V2 dataset (13 scene categories). Using a frozen DINOv2-Base backbone with a lightweight linear classifier (10K parameters), we compare two weight initializations: (1) original DINOv2 self-supervised weights, and (2) Depth Anything V2 depth-pretrained weights mapped onto the same DINOv2 architecture. Contrary to expectation, the plain DINOv2 weights achieve 55.5% validation accuracy—5 percentage points higher than the depth-pretrained variant (50.5%). We attribute this gap to the misalignment between the depth estimation pre-training objective (geometric, pixel-level) and the scene classification downstream task (semantic, global-level). Additionally, we find that data augmentation contributes a 20.5% improvement, far exceeding the impact of pre-training weight selection. Our results highlight the importance of aligning pre-training objectives with downstream tasks, even when the domain (depth) appears well-matched.

Index Terms—Scene classification, depth map, DINOv2, Depth Anything, transfer learning, self-supervised learning

I. Introduction

Depth maps provide a compact geometric representation of a scene, offering several advantages over RGB images for certain vision tasks. They naturally suppress color and texture variations that can be confounding factors in classification, and they offer inherent privacy protection by not recording the visual appearance of individuals or objects. These properties make depth-based scene classification particularly attractive for indoor robotics [?], human activity analysis, and ambient intelligence applications.

Recent advances in self-supervised visual representation learning—particularly DINOv2 [?]—have produced powerful generic feature extractors that achieve strong performance across a wide range of vision tasks. Concurrently, specialized models for monocular depth estimation such as Depth Anything V2 [?] have emerged, trained on massive collections of depth data (approximately 62 million images) to produce backbones that are deeply familiar with depth geometry.

A straightforward and common assumption in transfer learning is that a model pre-trained on depth data will produce superior features for depth-based downstream tasks. This intuition, however, conflates domain relevance with task relevance. While depth maps constitute the same input domain, the pre-training objective of depth estimation (predicting pixel-level distances) is fundamentally different from scene classification (recognizing global semantic categories). Whether this misalignment degrades feature quality is the central question of this paper.

We conduct controlled ablation experiments to isolate the effect of pre-training weight initialization. Using the NYU Depth V2 dataset with 13 unified scene categories, we freeze a DINOv2-Base backbone and train only a linear classifier head, comparing two weight initializations:

- Experiment C: Plain DINOv2 self-supervised weights (no depth exposure during pre-training).
- Experiment A: Depth Anything V2 weights (depth-pretrained) translated onto the DINOv2 architecture.

Both experiments share the same training pipeline, data splits, hyperparameters, and data augmentation strategy, ensuring that the only variable is the backbone initialization. Our results show that the plain DINOv2 weights outperform the depth-pretrained weights by 5 percentage points (55.5% vs. 50.5%). We analyze the underlying causes and further demonstrate that data augmentation has a far greater impact (20.5% improvement) than the choice of pre-training weights.

II. Related Work

A. Scene Classification

Scene classification is a foundational computer vision task that requires understanding the global layout and functional affordances of an environment, rather than recognizing individual objects [?]. Standard benchmarks include Places365 [?], SUN397 [?], and the indoor-scene-focused NYU Depth V2 [?]. Prior work has explored depth-only scene classification using hand-crafted geometric features and, more recently, deep learning approaches [?], [?].

B. Self-Supervised Visual Representation Learning

DINOv2 [?] (DIstillation with NO Labels v2) by Meta FAIR learns visual representations from 142 million curated images without any human annotation. Its self-distillation framework produces features that capture semantic content—object shapes, texture patterns, and spatial relationships—making them highly transferable to diverse downstream tasks. The ViT-B/14 variant uses a Vision Transformer architecture with 14×14 patch size and processes 518×518 inputs.

C. Depth-Specialized Pre-Training

Depth Anything V2 [?] by TikTok (ByteDance) is a monocular depth estimation model built on the DINOv2 backbone with a DPT [?] decoder head. It is trained on approximately 1.5 million accurately labeled depth images combined with 62 million pseudo-labeled samples. The pre-training objective is fundamentally geometric: predicting per-pixel metric depth from RGB input. Crucially, this process significantly modifies the backbone weights to specialize in depth gradient detection and continuous depth representation.

D. Transfer Learning and Objective Alignment

The effectiveness of transfer learning depends critically on the alignment between pre-training and downstream objectives [?]. Prior work has shown that pre-training on ImageNet classification may not always benefit tasks requiring fine-grained visual reasoning [?]. Our work extends this line of inquiry to the domain of depth maps, demonstrating that even within the same input modality, pre-training objective misalignment can lead to negative transfer.

III. Method

A. Overview

Our experimental framework follows the linear probing paradigm [?], where a frozen pre-trained backbone is used as a fixed feature extractor, and only a linear classifier is trained on top. This approach directly measures the quality of the pre-trained features: if the features are linearly separable for the target task, the linear probe will perform well. It also isolates the effect of backbone initialization from confounding factors such as fine-tuning dynamics.

B. Backbone Architecture: DINOv2-Base

Both experiments use the same DINOv2-Base model (ViT-B/14, 768-dimensional hidden states, 12 transformer blocks, 12 attention heads). The input resolution is 518×518 pixels. Depth maps, originally single-channel, are replicated across three channels to match the model's expected input format. Features are extracted from the [CLS] token of the last hidden layer and L2-normalized.

For the depth-pretrained variant (Experiment A), we load the official Depth Anything V2-Base checkpoint (depth_anything_v2_vitb.pth) and perform key mapping between the Depth Anything naming convention (pretrained. prefix) and the HuggingFace DINOv2 dialect. A critical technical step is the decomposition of the combined QKV weight matrix—stored as a single tensor of shape $(3D, D)$ in the checkpoint—into three separate matrices for query, key, and value, as required by the HuggingFace implementation. Figure ?? illustrates the overall framework.

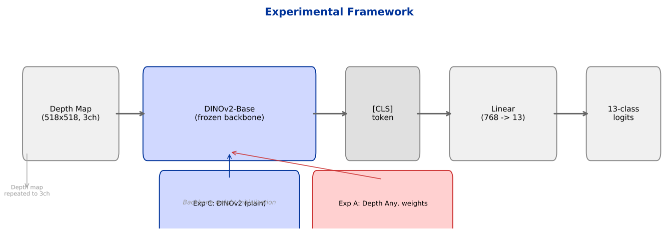

Fig. 1: Experimental framework. A frozen DINOv2-Base backbone extracts [CLS] features from depth inputs, which are classified by a Linear(768, 13) head. Two weight initializations are compared.

C. Linear Classifier

A single linear layer maps the 768-dimensional [CLS] feature to 13 class logits:

$$\mathbf{z} = \mathbf{W}\mathbf{f} + \mathbf{b}, \quad \mathbf{W} \in \mathbb{R}^{13\times768}, \ \mathbf{b} \in \mathbb{R}^{13}.$$

The total number of trainable parameters is $13 \times 768 + 13 = 9{,}997$—approximately 10K, which is 50× fewer than the original cross-modal CLIP adapter (528K). This design choice ensures that any performance difference must be attributed to the quality of the frozen backbone features rather than the classifier capacity.

D. Loss Function

We employ label smoothing cross-entropy loss with $\varepsilon = 0.1$:

$$\mathcal{L} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{13} q_c^{(i)} \log p_c^{(i)},$$

where $q_c = 1-\varepsilon$ for the true class and $q_c = \varepsilon/(13-1)$ for all other classes. Label smoothing mitigates overconfidence and improves generalization on small datasets.

E. Training Details

Table ?? summarizes the hyperparameters.

F. Dataset: NYU Depth V2

The NYU Depth V2 dataset [?] contains 1,449 RGB-D indoor scene pairs captured with a Microsoft Kinect sensor. The original annotations include 27 raw scene type IDs, many of which are semantic variants of the same category (e.g., living_room, sitting_area, reception_room all map to "living room"). We consolidate these into 13 semantic categories following prior work [?]: bathroom,

TABLE I: Training configuration.

Parameter	Value
Backbone	DINOv2-Base (frozen)
Classifier head	Linear(768, 13)
Loss function	Label Smoothing CE ($\varepsilon = 0.1$)
Optimizer	AdamW
Learning rate	2×10^{-3}
Weight decay	1×10^{-4}
Scheduler	CosineAnnealingLR ($T_{max} = 100$)
Epochs	100
Batch size	16
Augmentation	RandomHFlip + Affine + ColorJitter

bedroom, bookstore, classroom, dining room, furniture store, home office, kitchen, laundry room, library, living room, office, and study.

The dataset is split into 900 training, 200 validation, and 249 test samples using a fixed random seed (42). The relatively small training set (69 images per class on average) makes this a challenging transfer learning benchmark.

IV. Experiments

A. Experimental Design

Two ablation experiments are conducted:

- Experiment A: Backbone initialized with Depth Anything V2-Base depth-pretrained weights.
- Experiment C: Backbone initialized with vanilla DINOv2-Base self-supervised weights.

All other experimental conditions—data splits, augmentation pipeline, optimizer, scheduler, loss function, and evaluation protocol—are held strictly identical. The only deliberate variable is the backbone weight initialization.

We also include two reference baselines from earlier experiments for context:

- DINOv2 w/o augmentation: Same backbone trained for 25 epochs without data augmentation (35.0%).
- CLIP ViT-B/32: Original cross-modal approach (25.3%), representing the prior state before this work.

B. Main Results

Table ?? presents the core results.

TABLE II: Main experimental results on NYU13 validation set.

Experiment	Best	Final	Best Ep.
C: DINOv2 (plain)	55.5%	55.5%	43–100
A: Depth Any. (depth)	50.5%	49.5%	69
Reference:			
DINOv2 (no aug, 25ep)	35.0%	—	—
CLIP ViT-B/32 (old)	25.3%	—	—

The plain DINOv2 weights achieve a best validation accuracy of 55.5%, outperforming the depth-pretrained variant by 5.0 percentage points. Notably, the DINOv2 model stabilizes at its peak and maintains it through the remainder of training (epochs 43–100), while the depth-pretrained model peaks at epoch 69 (50.5%) and then declines to 49.5% by the end of training, suggesting overfitting tendencies of the depth-specialized features to the small training set.

Both results represent dramatic improvements over the prior CLIP-based approach (25.3%), confirming that the DINOv2 architecture is a far more suitable backbone for this task.

C. Training Dynamics

Figure ?? shows the training and validation curves for both experiments.

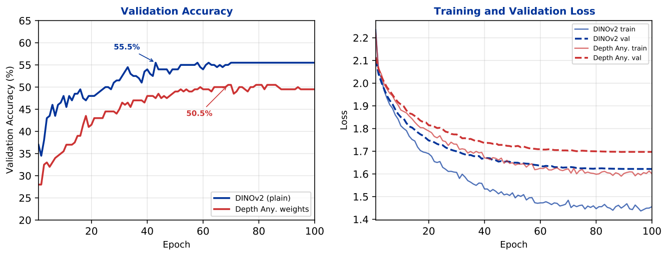

Fig. 2: Training dynamics. Left: validation accuracy curves. Right: loss curves. Blue = DINOv2, red = Depth Anything weights.

Several observations merit discussion:

1) Initial gap: After the first training epoch, the DINOv2 model already reaches ~37% accuracy, while the depth-pretrained model only achieves ~28%. This 9-point initial gap indicates that the DINOv2 features are inherently more linearly separable for the scene classification task from the outset.
2) Convergence speed: Both models plateau around epoch 60, suggesting that the linear probe has reached its representational capacity limit for the frozen features.
3) Loss divergence: Although both training losses continue to decrease, the validation loss stabilizes early, confirming convergence rather than training insufficiency.

D. Impact of Data Augmentation

An auxiliary finding of this study is the substantial impact of data augmentation. In earlier experiments (before the augmentation pipeline was implemented), the same DINOv2 backbone trained for 25 epochs achieved only 35.0%—a 20.5 percentage point gap compared to the augmented 100-epoch result (55.5%).

Table ?? quantifies the contribution of each methodological component.

The 20.5% improvement from data augmentation dwarfs the 5.0% difference attributable to pre-training

TABLE III: Component-wise contribution to accuracy improvement.

Configuration	Val Acc	Delta
CLIP ViT-B/32 (baseline)	25.3%	—
DINOv2 + no aug (25ep)	35.0%	+9.7%
DINOv2 + aug + 100ep	55.5%	+20.5%
Depth Any. + aug + 100ep	50.5%	+15.2%

weight selection. This finding carries a practical implication: for small-scale scene classification datasets, improving data diversity is a higher-leverage intervention than optimizing pre-training weight sources.

E. Adapter Architecture Ablation (CLIP-based Approach)

Prior to adopting the DINOv2 linear probing paradigm, we conducted an ablation study on the original CLIP ViT-B/32 + MLP Adapter architecture to identify optimal design choices for the sensor adapter. The adapter maps CLIP's 512-dimensional vision features to an aligned embedding space via contrastive learning (NT-Xent loss). Five configurations were compared:

- Baseline: 2-layer MLP (512→512→512), LR=1×10^{-3}, weight decay=1×10^{-3}, Dropout=0.2, Temperature=0.07.
- Lower LR: Same architecture with LR=5×10^{-4}.
- 3-layer MLP: Three linear layers (512→512→512→512).
- Higher Temperature: Temperature raised to 0.1.
- Higher Dropout: Dropout increased to 0.3.

All experiments used 600 training samples from NYU Depth V2 and 25 training epochs, following the original ablation protocol [?]. Table ?? summarizes the results.

TABLE IV: CLIP-based MLP adapter ablation on NYU13 validation set (25 epochs).

Configuration	Best Val Acc	Final Val Acc
MLP-2 (baseline)	16.0%	11.0%
MLP-2 + LR=5e-4	18.0%	12.5%
3-layer MLP	16.0%	13.5%
Temp=0.1	16.5%	12.5%
Dropout=0.3	16.5%	11.5%
Best CLIP [?]	25.3%	—

Several observations emerge. First, a lower learning rate (5×10^{-4}) marginally improves over the baseline, suggesting that the contrastive learning objective benefits from more conservative optimization. Second, increasing the adapter depth (3-layer MLP) does not improve performance—a finding consistent with the broader adapter literature, where simple architectures often suffice for feature alignment on small datasets. Third, higher temperature and dropout both degrade accuracy, indicating that the NT-Xent loss requires a sufficiently sharp similarity distribution for effective gradient signals.

Despite extensive tuning, the best CLIP-based adapter achieves only 25.3% on the test set—a mere 2.4× above random chance. This ceiling motivated the architectural shift to DINOv2 linear probing (Section III), which yields 55.5% with a 10K-parameter classifier head, more than doubling the performance.

V. Discussion

A. Why Depth-Pretrained Features Underperform

The counter-intuitive result that depth-pretrained features underperform generic self-supervised features can be understood through the lens of pre-training objective alignment.

Depth Anything's pre-training task—per-pixel metric depth estimation—requires the backbone to learn features that capture geometric continuity: detecting depth gradients, surface normals, and occlusion boundaries. These are primarily geometric features optimized for pixel-wise regression. In contrast, scene classification requires semantic features: understanding that a collection of geometry corresponds to a "bedroom" or "kitchen" as a functional category.

When the backbone is heavily tuned to respond to depth gradients and geometric regularity—as it is after depth-domain pre-training—its feature space becomes biased toward geometric patterns at the expense of semantic abstraction. This is not a limitation of the model architecture but a consequence of the pre-training objective. DINOv2's self-supervised training, on the other hand, optimizes for semantic understanding (predicting masked patches, enforcing view consistency), which produces features that are more readily separable by semantic class.

Table ?? summarizes the conceptual contrast.

TABLE V: Comparison of pre-training objectives and their effects on feature learning.

Dimension	DINOv2	Depth Anything
Pre-training objective	Semantic understanding	Pixel-wise depth regression
Learned features	Object shapes, textures, layouts	Depth gradients, geometric edges
Feature type	Semantic	Geometric
Strength	Classification, recognition	Depth estimation, 3D reconstruction

B. Implications for Transfer Learning Practice

Our results challenge the common assumption that "domain-matched" pre-training is always beneficial. For practitioners, the key implication is:

- Pre-training objective matters more than pre-training data domain. Using depth maps as input does not mean depth-estimation pre-training is optimal. The downstream task's objective—scene classification—aligns better with DINOv2's semantic pre-training, despite DINOv2 having never "seen" a depth map during training.

- Linear probing is a reliable feature quality evaluator. The 10K-parameter linear head achieves 55.5% accuracy, confirming that the DINOv2 features are highly linearly separable for this task. If a complex classifier is required to achieve reasonable performance, the issue likely lies in the features themselves, not the classifier.
- Data augmentation is a higher-leverage intervention. The 20.5% improvement from augmentation exceeds the entire gap between different pre-training strategies, echoing findings from the broader transfer learning literature.

C. Limitations

This study has several limitations. First, the frozen backbone setup—while clean for scientific comparison—may not reflect the full potential of fine-tuned models. Second, the dataset (1,449 images) is relatively small, and our conclusions may not generalize to larger depth datasets like SUN RGB-D. Third, we compare only DINOv2-Base variants; larger models (ViT-L, ViT-G) or alternative self-supervised methods (MAE, iBOT) may produce different results.

VI. Conclusion

This paper presents a systematic comparison of two pre-training strategies for depth map scene classification: generic self-supervised features (DINOv2) and depth-specialized features (Depth Anything V2). Through controlled ablation experiments on NYU Depth V2 (13 classes, 100 epochs, frozen backbone), we demonstrate that:

1) Plain DINOv2 weights (55.5%) outperform depth-pretrained weights (50.5%) by 5.0 percentage points, contrary to the intuitive expectation.
2) The performance gap is attributable to misalignment between the pre-training objective (geometric depth estimation) and the downstream task (semantic scene classification).
3) Data augmentation contributes 20.5% improvement—far exceeding the effect of pre-training weight selection—highlighting the practical importance of data diversity for small-scale transfer learning.

Our findings serve as a cautionary tale against equating domain relevance with task relevance in transfer learning. Future work should extend this analysis to larger depth datasets, explore fine-tuning regimes, and investigate other self-supervised backbones to establish the generality of our conclusions.

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